

Survival analysis with competing risks

Janne Pitkäniemi

Finnish Cancer Registry

Tampere University

Statistical Practice in Epidemiology (2026, Tartu)

Points to be covered

1. Survival or time to event data & censoring.
2. Competing risks: event-specific cumulative incidences & hazards.
3. Kaplan–Meier and Aalen–Johansen estimators.
4. Regression modelling of hazards: Cox model.
5. Packages `survival`, `mstate`, `Epi`, `(cmprisk)`.
6. Functions `Surv()`, `survfit()`, `plot.survfit()`, `coxph()`.

Survival time – time to event

Time spent (lex.dur) in a given **state** (lex.Cst) from its beginning till a certain *endpoint* or *outcome event* (lex.Xst) or *transition* occurs, changing the state to another.

Examples of such times and outcome events:

- ▶ lifetime: birth $\rightarrow$ death,
- ▶ duration of marriage: wedding $\rightarrow$ divorce,
- ▶ healthy exposure time:
start of exposure $\rightarrow$ onset of disease,
- ▶ clinical survival time:
diagnosis of a disease $\rightarrow$ death.

Ex. Survival of 338 oral cancer patients

Important variables:

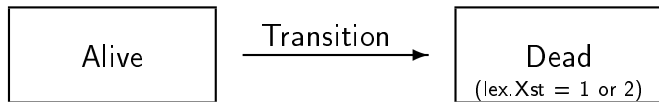
- ▶ `time` = duration of patientship from diagnosis (**entry**) till death (`death`) or censoring (`Alive`), (`lex.Cst` is (`Alive`))
- ▶ `event` = indicator for the outcome and its observation at the end of follow-up (**exit**):
0 = censoring,
1 = death from oral cancer

Special features:

- ▶ Two possible endpoints
- ▶ Censoring – incomplete observation of the survival time.

Set-up of classical survival analysis

- ▶ **Two-state model:** only one type of event changes the initial state.
- ▶ Major applications: analysis of lifetimes since birth and of survival times since diagnosis of a disease until death from any cause.



- ▶ **Censoring:** Death and final lifetime not observed for some subjects due to emigration or closing the follow-up while they are still alive

Distribution concepts: hazard function

The **hazard rate** or **intensity** function $\lambda(t)$

$$\lambda(t) = \lim_{\Delta \rightarrow 0} \frac{P(t < T \leq t + \Delta | T > t)}{\Delta},$$

- ▶ the conditional probability that the event occurs in a short interval $(t, t + \Delta]$, given that it does not occur before t , divided by interval length which gets smaller Δ

During a short interval

risk of event $\approx$ hazard $\times$ interval length

Distribution concepts: survival and cumulative hazard functions

Survival function

$$S(t) = P(T > t),$$

= probability of avoiding the event at least up to t (the event occurs only after t).

The **cumulative hazard** (or integrated intensity):

$$\Lambda(t) = \int_0^t \lambda(u) du$$

one to one connections between the functions, cumulative incidence function, survival and cumulative hazard:

$$F(t) = 1 - S(t)$$

$$S(t) = \exp\{-\Lambda(t)\}$$

Observed data on survival times

For individuals $i = 1, \dots, n$ let

T_i = time to outcome event,

U_i = time to censoring.

Censoring is assumed **noninformative**, *i.e.* independent from occurrence of events.

We observe

$y_i = \min\{T_i, U_i\}$, *i.e.* the exit time, and

$\delta_i = 1_{\{T_i < U_i\}}$, indicator (1/0) for the outcome event occurring first, before censoring.

Censoring must properly be taken into account in the statistical analysis.

Approaches for analysing survival time

- ▶ **Parametric model** (like Weibull, gamma, etc.) on hazard rate $\lambda(t) \rightarrow$ Likelihood:

$$L = \prod_{i=1}^n \lambda(y_i)^{\delta_i} S(y_i)$$

- ▶ **Piecewise constant rate model** on $\lambda(t)$
 - see Bendix's lecture on time-splitting (Poisson likelihood).
- ▶ **Non-parametric** methods, like Kaplan–Meier (KM) estimator of survival curve $S(t)$ and Cox proportional hazards model on $\lambda(t)$.

R package survival

Tools for analysis with one outcome event.

- ▶ `Surv(time, event) -> sobj`
creates a **survival object** `sobj` assuming that all start at 0, containing pairs (y_i, δ_i) ,
- ▶ `Surv(entry, exit, event) -> sobj2`
creates a survival object from entry and exit times,
- ▶ `survfit(sobj ~ x) -> sfo`
creates a **survfit** object `sfo` containing KM or other non-parametric estimates (also from a fitted Cox model),
- ▶ `plot(sfo), plotCIF(sobj)`
plot method for survival curves and related graphs,
- ▶ `coxph(sobj ~ x1 + x2)`
fits a Cox model with covariates `x1` and `x2`.
- ▶ `survreg()` – parametric survival models.

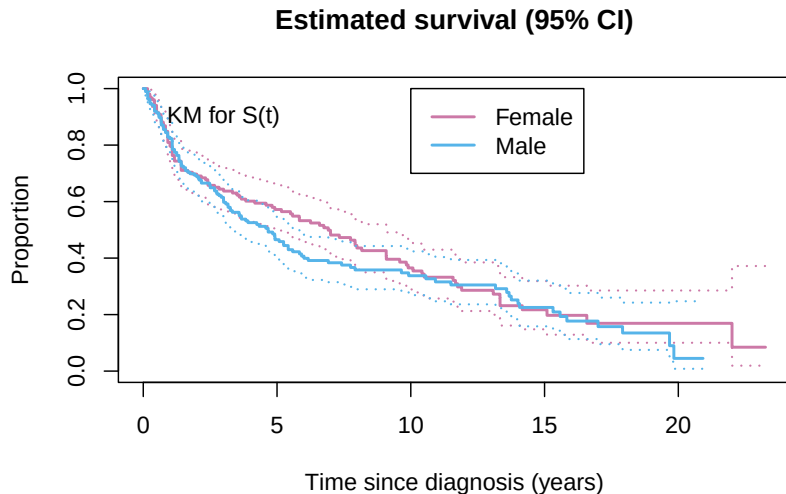
Ex. Oral cancer data (cont'd)

```
> suob <- Surv(orca$time, 1*(orca$event > 0) )
> suob[1:7] # + indicates censored observation
[1] 5.081+ 0.419 7.915 2.480 2.500 0.167 5.925+
> km1 <- survfit( suob ~ 1, data = orca)
> km1 # brief summary
Call: survfit(formula = suob ~ 1, data = orca)
```

```
      n events median 0.95LCL 0.95UCL
[1,] 338      229  5.42    4.33    6.92
> summary(km1) # detailed KM-estimate
Call: survfit(formula = suob ~ 1, data = orca)
```

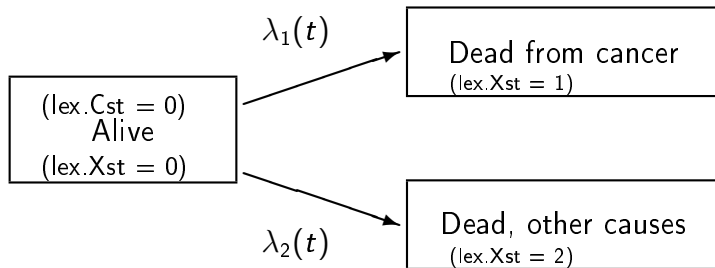
time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95% CI
0.085	338	2	0.9941	0.00417	0.9859	1.000
0.162	336	2	0.9882	0.00588	0.9767	1.000
0.167	334	4	0.9763	0.00827	0.9603	0.993
0.170	330	2	0.9704	0.00922	0.9525	0.989
0.246	328	1	0.9675	0.00965	0.9487	0.987
0.249	327	1	0.9645	0.01007	0.9450	0.984
0.252	326	3	0.9556	0.01120	0.9339	0.978
0.329	323	1	0.9527	0.01155	0.9303	0.976
0.334	322	1	0.9497	0.01189	0.9267	0.973
0.413	321	1	0.9467	0.01221	0.9231	0.971

Oral cancer: Kaplan-Meier estimates



Competing risks model: causes of death

- ▶ Often the interest is focused on the risk or hazard of dying from one specific cause.
- ▶ That cause may eventually not be realized, because a **competing cause** of death hits first.



- ▶ Generalizes to several competing causes.

Competing events & competing risks

In many epidemiological and clinical contexts there are competing events that may occur before the target event and remove the person from the population at risk for the event, e.g.

- ▶ *target event*: occurrence of endometrial cancer, *competing events*: hysterectomy or death.
- ▶ *target event*: relapse of a disease (ending the state of remission), *competing event*: death while still in remission.
- ▶ *target event*: divorce, *competing event*: death of either spouse.

Event-specific quantities

Cumulative incidence function (CIF) or

$$F_c(t) = P(T \leq t \text{ and } C = c), \quad c = 1, 2,$$

From these one can recover

- ▶ $F(t) = \sum_c F_c(t)$, CDF of event-free survival time T , *i.e.* cumulative risk of any event by t .
- ▶ $S(t) = 1 - F(t)$, **event-free survival function**, *i.e.* probability of avoiding all events by t , but $S(t) \neq F_1(t) + F_2(t)$

We have lost one-to-one connection between cause-specific hazards and cumulative incidence (earlier slide).

Event-specific quantities (cont'd)

Event- or cause-specific hazard function

$$\begin{aligned}\lambda_c(t) &= \lim_{\Delta \rightarrow 0} \frac{P(t < T \leq t + \Delta \text{ and } C = c \mid T > t)}{\Delta} \\ &= \frac{f_c(t)}{1 - F(t)}\end{aligned}$$

CIF = risk of event c over risk period $[0, t]$ in the presence of competing risks, also obtained

$$F_c(t) = \int_0^t \lambda_c(v) S(v) dv, \quad c = 1, 2,$$

More on the technical definitions of relevant quantities:

<http://bendixcarstensen.com/AdvCoh/papers/fundamentals.pdf>

Warning of “net risk” and “cause-specific survival”

- ▶ The “**net risk**” of outcome c by time t , assuming hypothetical elimination of competing risks, is often defined as

$$F_1^*(t) = 1 - S_1^*(t) = 1 - \exp\{-\Lambda_1(t)\} \neq S(t)$$

- ▶ In clinical survival studies, function $S_1^*(t)$ is often called “**cause-specific survival**”, or “**net survival**”
- ▶ Yet, these *-functions, $F_1^*(t)$ and $S_1^*(t)$, lack proper probability interpretation when competing risks exist.
- ▶ Hence, their use should be viewed critically (Andersen & Keiding, *Stat Med*, 2012)

Analysis with competing events

Let U_i = censoring time, T_i = time to first event, and C_i = variable for event 1 or 2. We observe

- ▶ $y_i = \min\{T_i, U_i\}$, i.e. the exit time, and
- ▶ $\delta_{ic} = 1_{\{T_i < U_i \text{ \& } C_i = c\}}$, indicator (1/0) for event c being first observed, $c = 1, 2$.

Non-parametric estimation of CIF

- ▶ Let $t_1 < t_2 < \dots < t_K$ be the K distinct time points at which any outcome event was observed,
- ▶ **Aalen-Johansen estimator** (AJ) for the cumulative incidence function $F(t)$ should be used

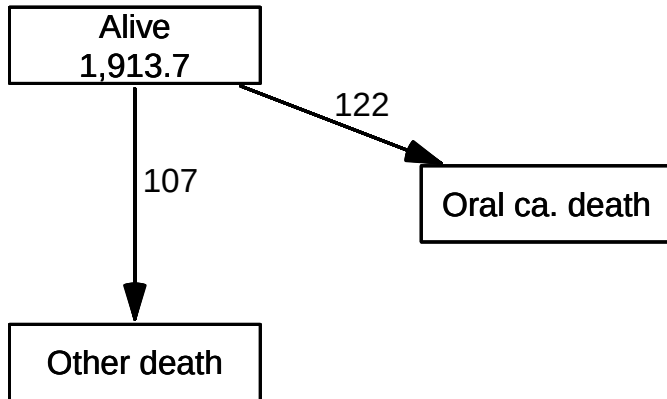
R tools for competing risks analysis

- ▶ `survfit( Surv(...,type="mstate") )` in Survival-package can be fitted for any transition of a multistate model and to obtain A-J estimates.
- ▶ Package `cmprsk` – `cuminc(ftime, fstatus, ...)` computes CIF-estimates, and can be compared in more than two samples. `plot.cuminc()` plots them.
- ▶ Package `Epi` – Lexis tools for multistate analyses
Will be advertised by Bendix!

Box diagram for transitions

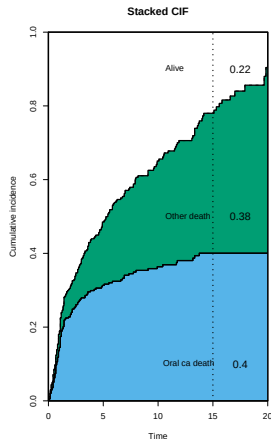
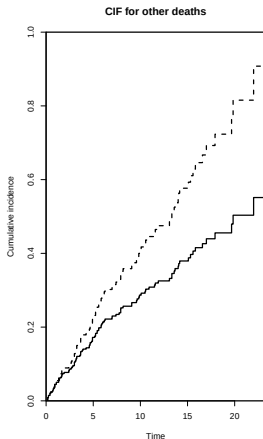
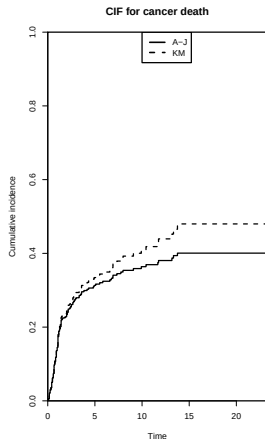
NOTE: entry.status has been set to "Alive" for all.

NOTE: entry is assumed to be 0 on the stime timescale.

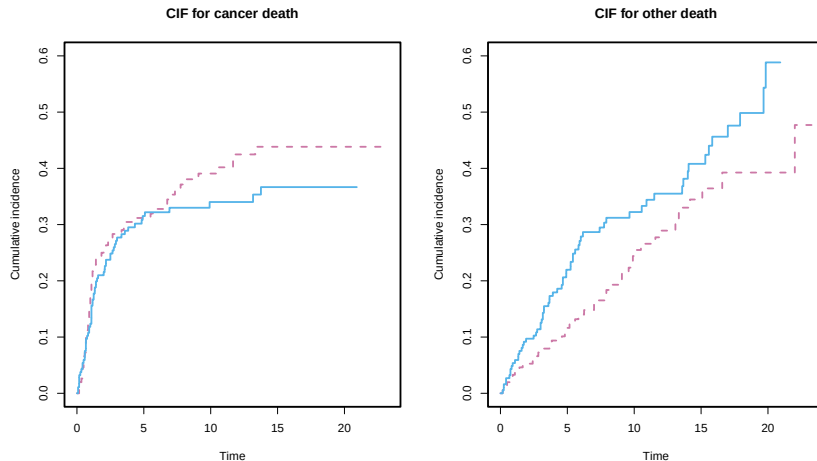


Ex. Survival from oral cancer

- ▶ AJ-estimates of CIFs (solid) for both causes.
- ▶ Naive KM-estimates of CIF (dashed) $>$ AJ-estimates
- ▶ CIF curves may also be stacked (right).



Ex. CIFs by cause in men and women



CIF for cancer higher in women (chance?) but for other causes higher in men (no surprise).

Regression models for time-to-event data

Regression models for hazards can be defined e.g. for

(a) hazards, multiplicatively:

$$\lambda_i(t) = \lambda_0(t; \alpha) r(\eta_i), \quad \text{where}$$

$\lambda_0(t; \alpha)$ = baseline hazard and

$r(\eta_i)$ = relative rate function, typically $\exp(\eta_i)$

(b) hazards, additively:

$$\lambda_i(t) = \lambda_0(t; \alpha) + \eta_i.$$

Relative hazards model or Cox model

In model (b), the baseline hazard $\lambda_0(t, \alpha)$ may be given a parametric form (e.g. Weibull) or a piecewise constant rate (exponential) structure.

Often a parameter-free form $\lambda_0(t)$ is assumed. Then

$$\lambda_i(t) = \lambda_0(t) \exp(\eta_i),$$

specifies the **Cox model** or the **semiparametric proportional hazards model**. $\eta_i = \beta_1 x_{i1} + \dots + \beta_p x_{ip}$ not depending on time.

Generalizations: **time-dependent** covariates $x_{ij}(t)$

PH model: interpretation of parameters

Present the model explicitly in terms of x 's and β 's.

$$\lambda_i(t) = \lambda_0(t) \exp(\beta_1 x_{i1} + \cdots + \beta_p x_{ip})$$

Consider two individuals, i and i' , having the same values of all other covariates except the j^{th} one.

The ratio of hazards is constant:

$$\frac{\lambda_i(t)}{\lambda_{i'}(t)} = \frac{\exp(\eta_i)}{\exp(\eta_{i'})} = \exp\{\beta_j(x_{ij} - x_{i'j})\}.$$

Thus $e^{\beta_j} = \text{HR}_j = \mathbf{hazard\ ratio}$ or relative rate associated with a unit change in covariate X_j .

Ex. Total mortality of oral ca. patients

Fitting Cox models with sex and sex + age.

```
> cm0 <- coxph( suob ~ sex, data = orca)
> ci.exp(cm0)
```

	exp(Est.)	2.5%	97.5%
sexMale	1.134004	0.8724905	1.473902

Total mortality in males is 13% higher in male than females, but not significant.

```
> cm0 <- coxph( suob ~ age+sex, data = orca)
> ci.exp(cm0)
```

	exp(Est.)	2.5%	97.5%
age	1.041914	1.030655	1.053296
sexMale	1.494305	1.139254	1.960009

The M/F contrast visible only after age-adjustment.(49% higher in males).

Predictions from the Cox model

- ▶ Individual survival *times* cannot be predicted but ind'l survival *curves* can.
PH model implies:

$$S_i(t) = [S_0(t)]^{\exp(\beta_1 x_{i1} + \dots + \beta_p x_{ip})}$$

- ▶ Having estimated β by partial likelihood, the baseline $S_0(t)$ is estimated by Breslow method
- ▶ From these, a survival curve for an individual with given covariate values is predicted.
- ▶ In R: `pred <- survfit(mod, newdata=...)` and `plot(pred)`, where `mod` is the fitted `coxph` object, and `newdata` specifies the covariate values. `newdata` is always needed for predictions.

Modelling with competing risks

Main options, providing answers to different questions.

- (a) Cox model for event-specific hazards $\lambda_c(t) = f_c(t)/[1 - F(t)]$, when *e.g.* the interest is in the biological effect of the prognostic factors on the fatality of the very disease that often leads to the relevant outcome.

- (b) **Fine–Gray model** for the hazard of the subdistribution $\gamma_c(t) = f_c(t)/[1 - F_c(t)]$ when we want to assess the impact of the factors on the overall cumulative incidence of event c .
 - Function `crr()` in package `cmprsk`.

SMR

Relate population mortality to the mortality of your "exposed" cohort

Let

- ▶ $\lambda(a)$ be the mortality in the cohort
- ▶ $\lambda_P(a)$ be the population mortality
- ▶ $\lambda_E(a)$ be the excess hazard of dying from the disease among cohort members
- ▶ SMR is the relative mortality in the cohort

$$\lambda(a) = \lambda_E(a) + \lambda_P(a) \text{ (excess mortality)}$$

$$\lambda(a) = SMR \times \lambda_P(a) \text{ (standardized mortality ratio)}$$